MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

NAND/SCHMITT, DIGITAL, KEY CODE JQM

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND/Schmitt module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.087 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be JQM.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/222-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

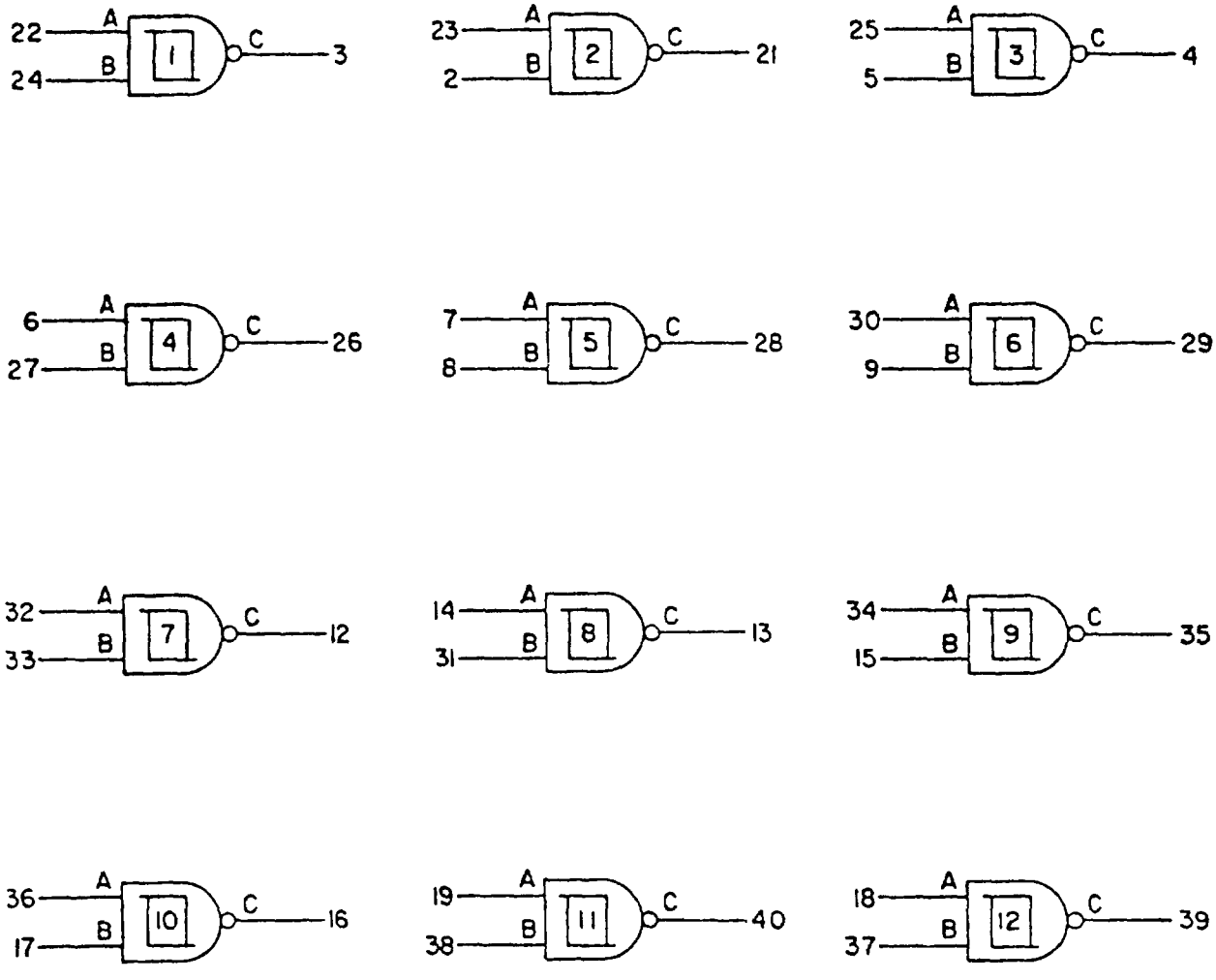
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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Pin 1 - V_{CC}(+5 V dc±10%)
Pin 10 - GND
Pins 11, 20(Not used)

* FIGURE 1. JQM module function diagram, NAND/Schmitt.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation, frame	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
A2,A3	Isolation, unused pins	III	X	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B12	V_{OUT} (low loaded)	IV	S	10	800	10	800	10	800	10	800	mV
C1-C24	Input breakdown current	V	S	0	1	0	1	0	1	0	1	mA
D1-D24	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E12	Output voltage, low level	VII	X	0	400	0	400	0	400	0	400	mV
F1-F24	Output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
G1-G24	Input current, low level, V_{IN} = 0 V	IX	X	0.40	1.32	0.40	1.32	0.40	1.32	0.40	1.32	mA
H1-H24	Input current, high level, V_{IN} = 2.4 V	X	X	0	40	0	40	0	40	0	40	µA
J1-J12	Output short circuit current	XI	X	18	55	18	55	18	55	18	55	mA
K1-K24	Input threshold voltage (V_{THL})	XII	X	1.5	2.0	1.5	2.0	1.5	2.0	1.5	2.0	V
L1-L24	Input threshold voltage (V_{TLH})	XII	X	0.6	1.1	0.6	1.1	0.6	1.1	0.6	1.1	V
M1-M24	Hysteresis (V_{THL} - V_{TLH})	XII	X	0.4	1.4	0.4	1.4	0.4	1.4	0.4	1.4	V
N1	Power supply current, low level	XIII	S	1	120	1	120	1	120	1	120	mA
N2	Power supply current, high level	XIII	S	1	72	1	72	1	72	1	72	mA
P1-P12	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	4	45	4	45	4	50	4	50	ns
P13-P24	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	4	55	4	55	4	60	4	60	ns
Q1-Q12	Transition time (t_{THL}), C_L = 50 pF	XIV	S	4	20	4	20	4	25	4	25	ns

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q13-Q24	Transition time (t_{THL}), C_L = 150 pF	XIV	Q	4	30	4	30	4	35	4	35	ns
R1-R12	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	5	30	5	30	5	33	5	33	ns
R13-R24	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	5	40	5	40	5	43	5	43	ns
S1-S12	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	5	30	5	30	5	33	5	33	ns
S13-S24	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	5	40	5	40	5	43	5	43	ns
T1-T12	Output crosstalk, low level	XVI	Q	0	1	0	1	0	1	0	1	V
T13-T24	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
U1-U12	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
V	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	mV
W	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
X1-X24	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through W shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition W, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through X shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through X shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/222-1 supersedes part number M28787/222. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	660 mW (all outputs in low level) 396 mW (all outputs in high level)
Supply voltage	5 V dc ±10 percent
Input threshold voltage Negative going (V_{TLH}) Positive going (V_{THL})	 600 mV minimum, 1.1 V maximum 1.5 V minimum, 2 V maximum
Input threshold margin	400 mV minimum

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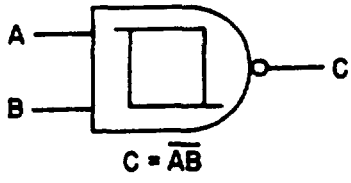
Output voltage	
Low level	400 mV maximum
High level	2.4 V minimum, V_{CC} maximum
Input current	
Low level (V_{IL} = 0 V)	-1.32 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Output drive	
Low level (V_{OUT} at 400 mV)	16 mA maximum
High level (V_{OUT} at 2.4 V)	-800 µA maximum

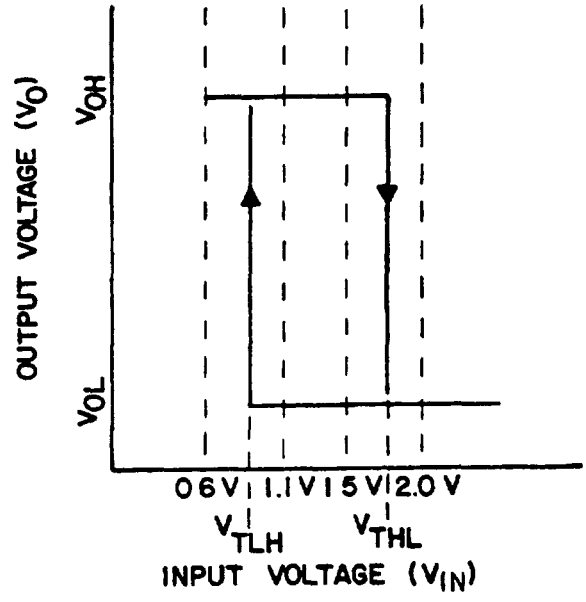
* 6.5 Module function. This module contains 12 independent 2-input Schmitt-trigger positive-NAND gates as illustrated on figure 1.

6.5.1 Input to output relationship. Each gate has the input to output relationship of table II for positive logic for a high level (H) defined as an input voltage greater than V_{THL} (max), a low level (L) defined as an input voltage less than V_{TLH} (min), and positive logic.

TABLE II. Input to output relationship.

Input		Output
L	L	H
L	H	H
H	L	H
H	H	L





6.5.1.1 Input threshold margin. Each input shall have a positive-going threshold voltage (V_{THL}) that is at least 400 mV greater than its negative-going threshold voltage (V_{TLH}).

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.5.3 Crosstalk relationship. The module and circuits have the following crosstalk relationships.

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6.5.3.1 Module crosstalk. When connected in the worst case mode of operation (all but one output switching simultaneously from a high level to a low level) the JQM module could exhibit substantial crosstalk. When the quiet output is a low level, there may be as much as 1 V positive-going spike coincident with the negative edge of the switching outputs.

6.5.3.2 Circuit crosstalk. Integrated circuits 2 and 3 can induce as much as 400mV of noise onto any IC circuit 1 output. Integrated circuits 1 and 3 can induce as much as 400 mV of noise onto an IC circuit 2 output. Integrated circuits 1 and 2 can induce as much as 400 mV noise onto any IC circuit 3 output.

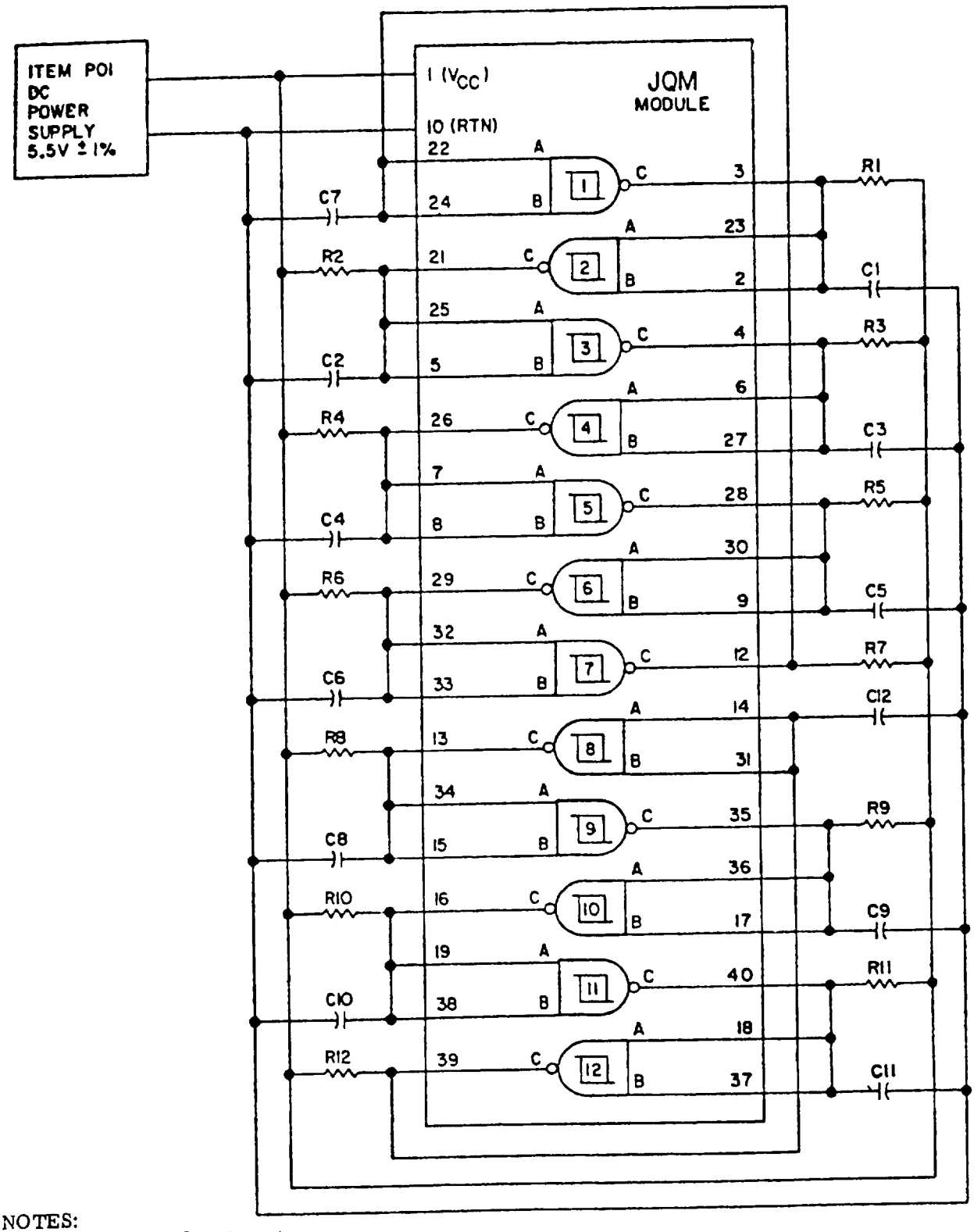
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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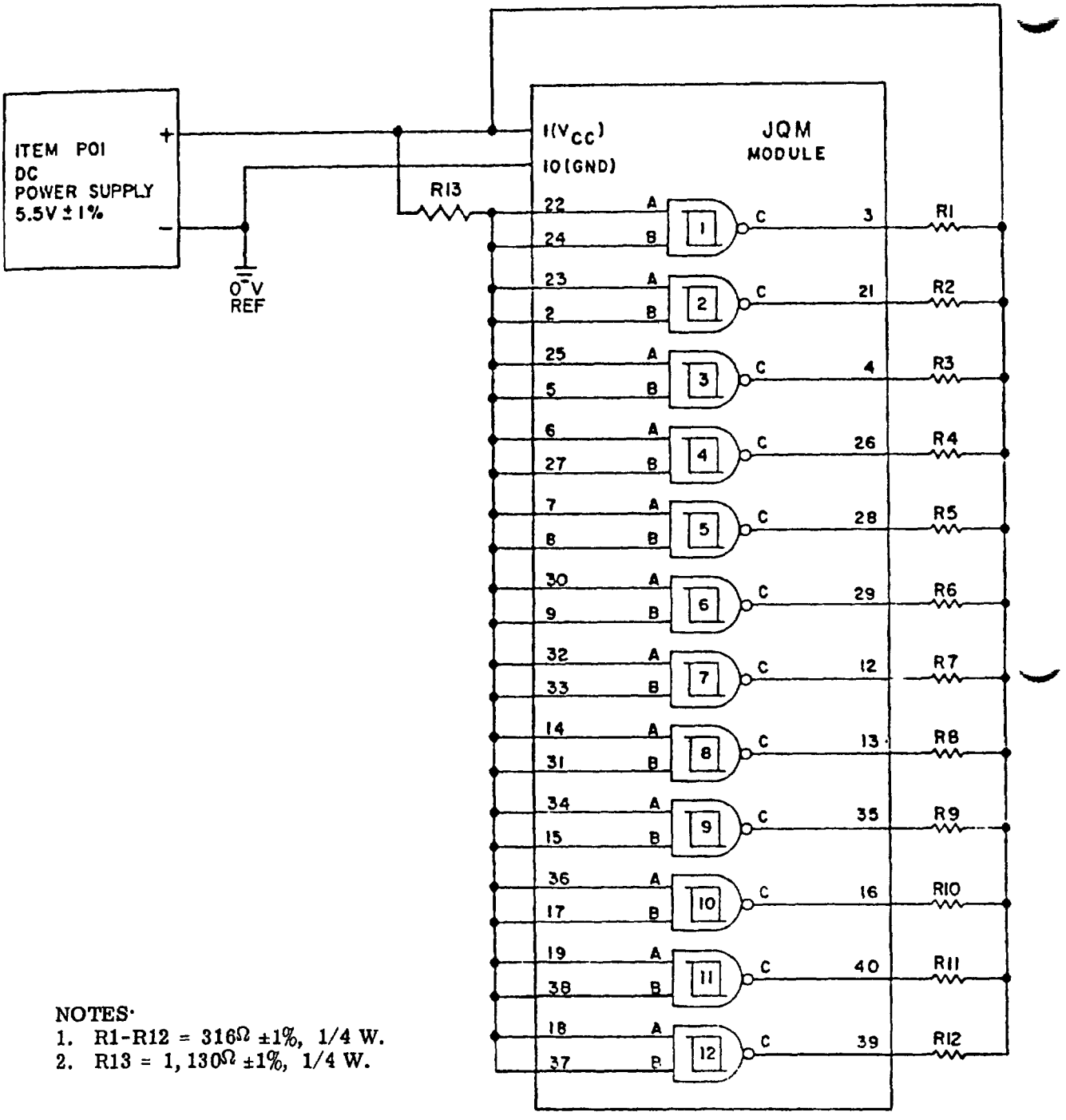
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NOTES:
1. R1-R12 = 316Ω ±1%, 1/4 W.
2. C1-C12 = 100 pF ±20%.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1-R12 = 316Ω ±1%, 1/4 W.
2. R13 = 1,130Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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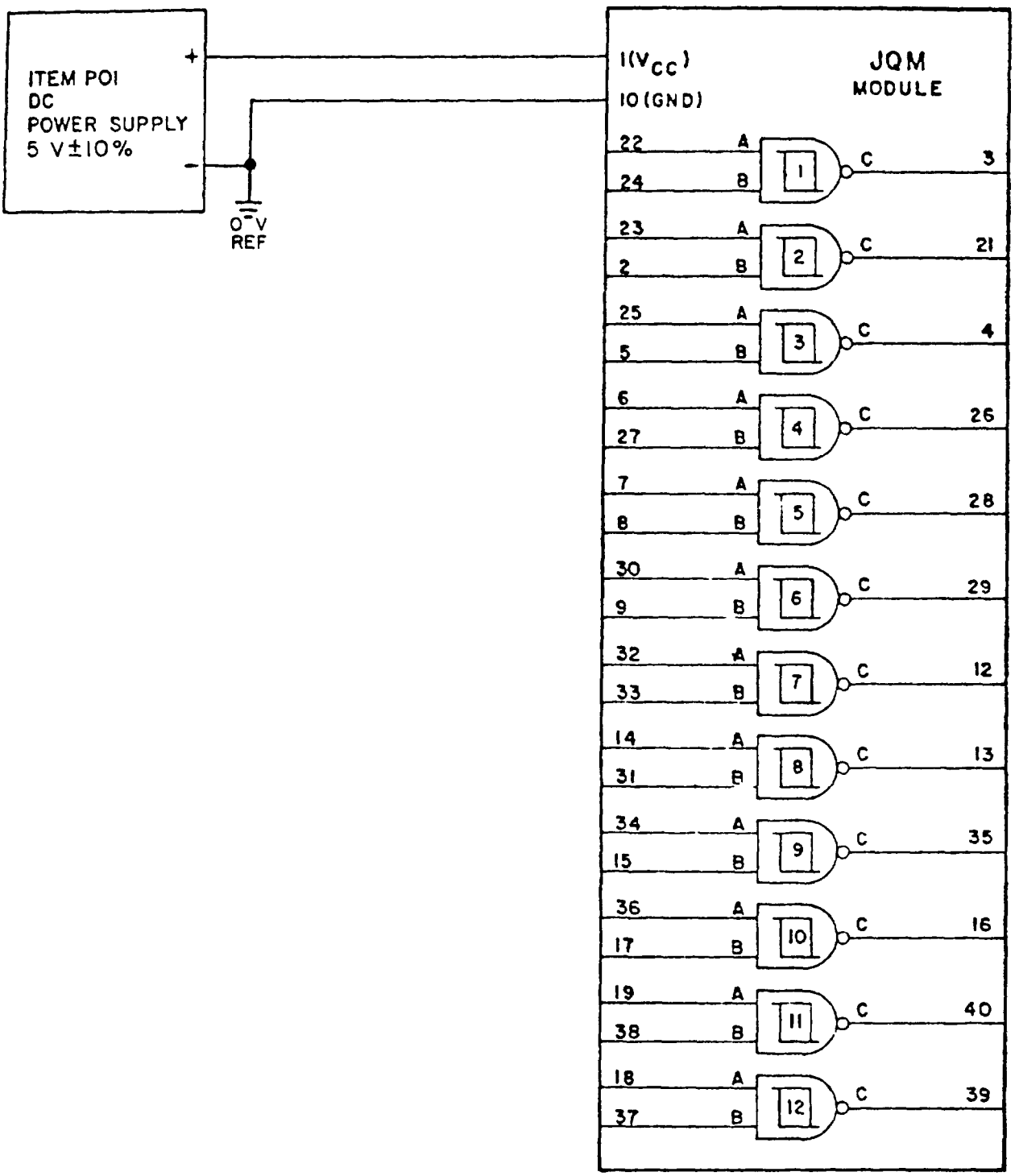


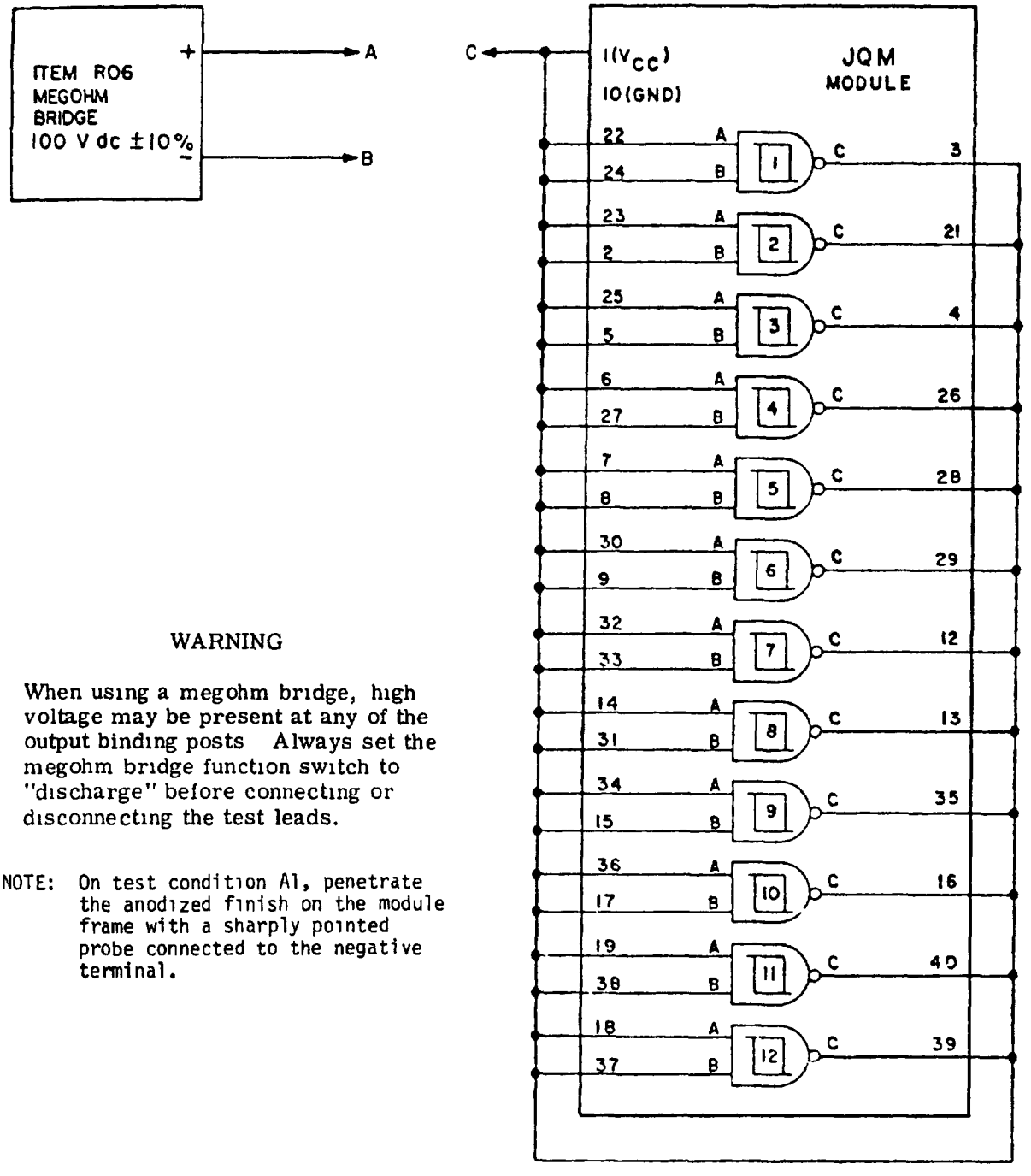
FIGURE 4 Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	C,11,20	Frame	10	∞	MΩ	5
A2	Isolation, unused pins	C	11,20	"	"	"	"
A3	Isolation, unused pins	11	20	"	"	"	"

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: On test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

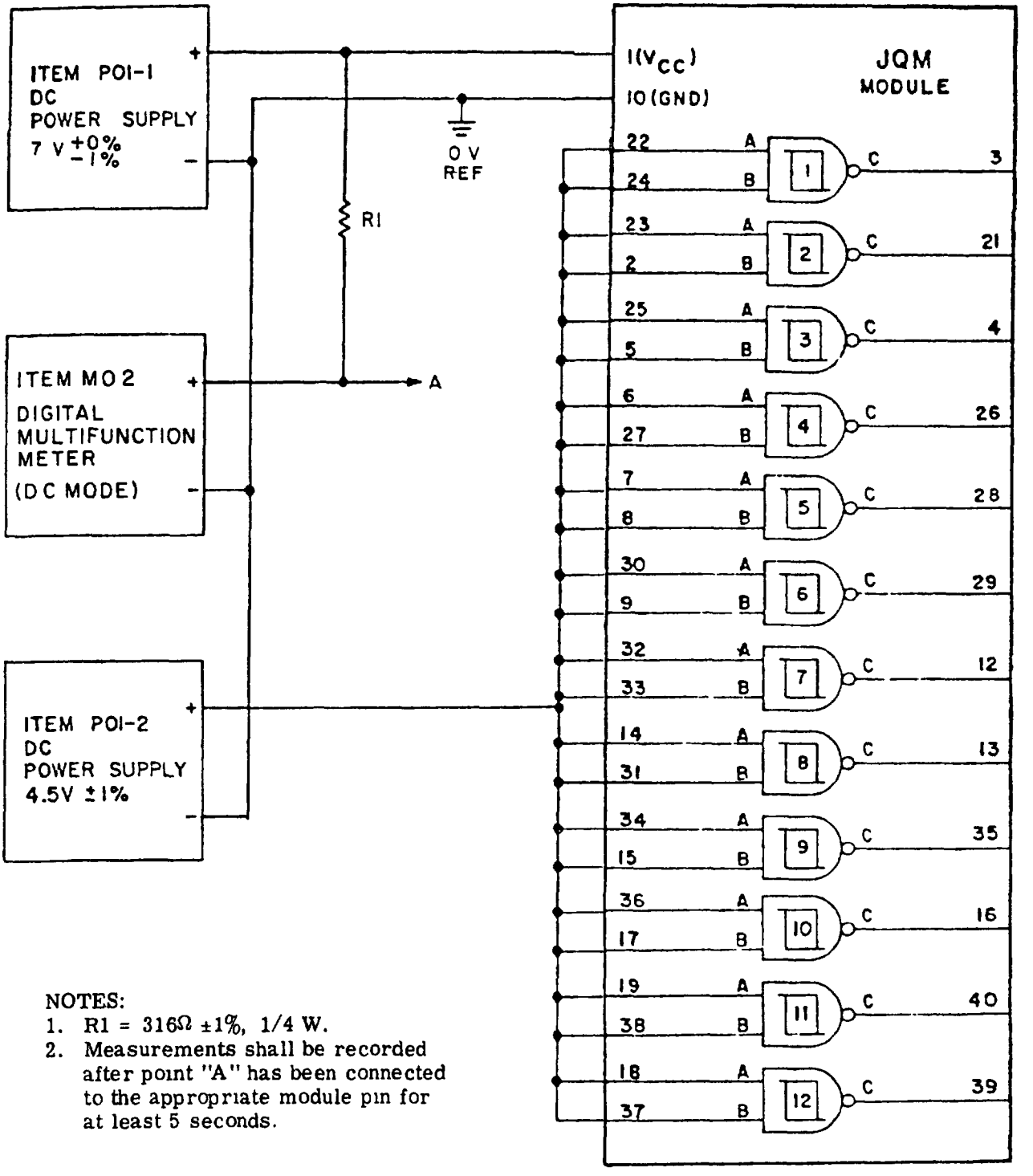
FIGURE 5. Isolation test circuit for table III.

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TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
B1	1C V_{OUT} (low loaded)	3	10	800	mV	6
B2	2C "	21	"	"	"	"
B3	3C "	4	"	"	"	"
B4	4C "	26	"	"	"	"
B5	5C "	28	"	"	"	"
B6	6C "	29	"	"	"	"
B7	7C "	12	"	"	"	"
B8	8C "	13	"	"	"	"
B9	9C "	35	"	"	"	"
B10	10C "	16	"	"	"	"
B11	11C "	40	"	"	"	"
B12	12C "	39	"	"	"	"

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NOTES:
1. R1 = 316Ω ±1%, 1/4 W.
2. Measurements shall be recorded after point "A" has been connected to the appropriate module pin for at least 5 seconds.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input breakdown current	22	2,5-9,14,15,17-19, 23-25,27,30-34,36-38	0	1	mA	7
C2	1B "	24	2,5-9,14,15,17-19,22, 23,25,27,30-34,36-38	"	"	"	"
C3	2A "	23	2,5-9,14,15,17-19,22, 24,25,27,30-34,36-38	"	"	"	"
C4	2B "	2	5-9,14,15,17-19, 22-25,27,30-34,36-38	"	"	"	"
C5	3A "	25	2,5-9,14,15,17-19, 22-24,27,30-34,36-38	"	"	"	"
C6	3B "	5	2,6-9,14,15,17-19, 22-25,27,30-34,36-38	"	"	"	"
C7	4A "	6	2,5,7-9,14,15,17-19, 22-25,27,30-34,36-38	"	"	"	"
C8	4B "	27	2,5-9,14,15,17-19, 22-25,30-34,36-38	"	"	"	"
C9	5A "	7	2,5,6,8,9,14,15,17-19, 22-25,27,30-34,36-38	"	"	"	"
C10	5B "	8	2,5-7,9,14,15,17-19, 22-25,27,30-34,36-38	"	"	"	"
C11	6A "	30	2,5-9,14,15,17-19, 22-25,27,31-34,36-38	"	"	"	"
C12	6B "	9	2,5-8,14,15,17-19, 22-25,27,30-34,36-38	"	"	"	"
C13	7A "	32	2,5-9,14,15,17-19,22- 25,27,30,31,33,34, 36-38	"	"	"	"
C14	7B "	33	2,5-9,14,15,17-19,22- 25,27,30-32,34,36-38	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C15	8A Input breakdown current	14	2,5-9,15,17-19,22-25, 27,30-34,36-38	0	1	mA	7
C16	8B "	31	2,5-9,14,15,17-19, 22-25,27,30,32-34, 36-38	"	"	"	"
C17	9A "	34	2,5-9,14,15,17-19, 22-25,27,30-33,36-38	"	"	"	"
C18	9B "	15	2,5-9,14,17-19,22-25, 27,30-34,36-38	"	"	"	"
C19	10A "	36	2,5-9,14,15,17-19, 22-25,27,30-34,37,38	"	"	"	"
C20	10B "	17	2,5-9,14,15,18,19, 22-25,27,30-34,36-38	"	"	"	"
C21	11A "	19	2,5-9,14,15,17,18, 22-25,27,30-34,36-38	"	"	"	"
C22	11B "	38	2,5-9,14,15,17-19, 22-25,27,30-34,36,37	"	"	"	"
C23	12A "	18	2,5-9,14,15,17,19, 22-25,27,30-34,36-38	"	"	"	"
C24	12B "	37	2,5-9,14,15,17-19, 22-25,27,30-34,36,38	"	"	"	"

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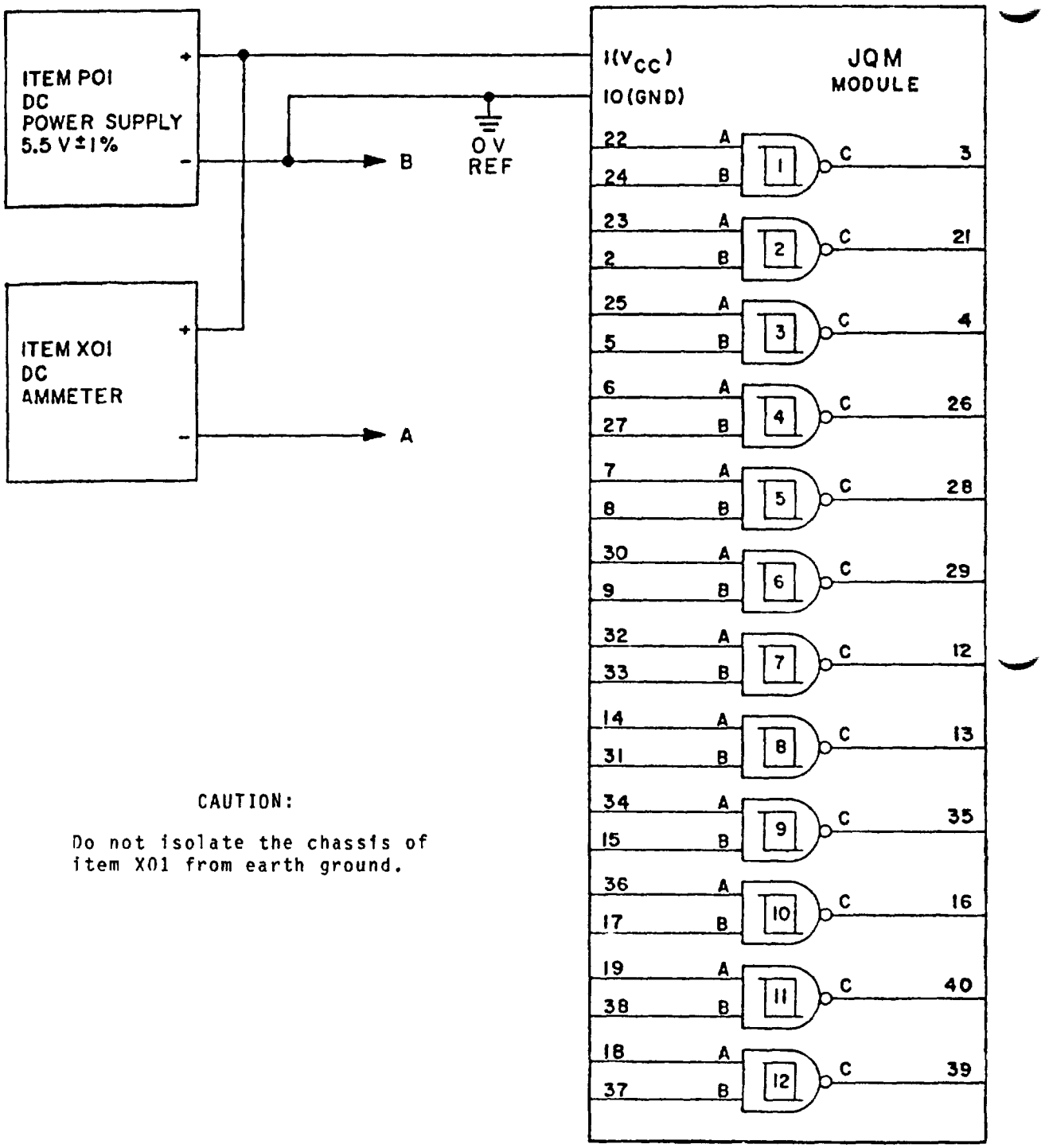


FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Input clamp diode voltage test.

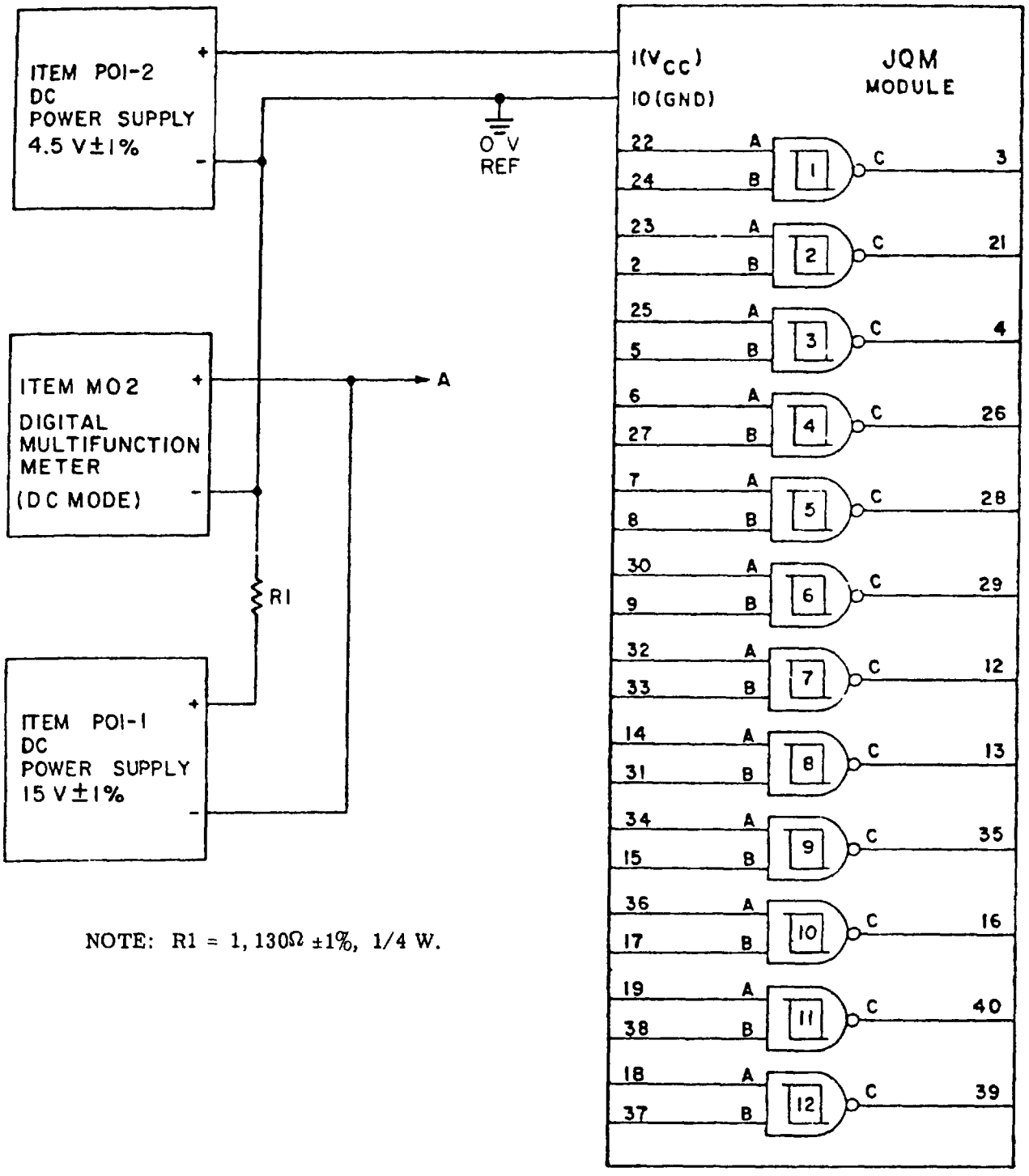
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D1	1A Input clamp diode voltage	22	-0.4	-1.5	V	8
D2	1B "	24	"	"	"	"
D3	2B "	2	"	"	"	"
D4	2A "	23	"	"	"	"
D5	3B "	5	"	"	"	"
D6	3A "	25	"	"	"	"
D7	4A "	6	"	"	"	"
D8	4B "	27	"	"	"	"
D9	5A "	7	"	"	"	"
D10	5B "	8	"	"	"	"
D11	6B "	9	"	"	"	"
D12	6A "	30	"	"	"	"
D13	7A "	32	"	"	"	"
D14	7B "	33	"	"	"	"
D15	8A "	14	"	"	"	"
D16	8B "	31	"	"	"	"
D17	9B "	15	"	"	"	"
D18	9A "	34	"	"	"	"
D19	10B "	17	"	"	"	"
D20	10A "	36	"	"	"	"
D21	11A "	19	"	"	"	"

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TABLE VI. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D22	11B Input clamp diode voltage	38	-0.4	-1.5	V	8
D23	12A "	18	"	"	"	"
D24	12B "	37	"	"	"	"

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NOTE: R1 = 1,130Ω ±1%, 1/4 W.

* FIGURE 8. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	1C Output voltage, low level	22,24	Open	3	0	400	mV	9
E2	2C "	2,23	"	21	"	"	"	"
E3	3C "	5,25	"	4	"	"	"	"
E4	4C "	6,27	"	26	"	"	"	"
E5	5C "	7,8	"	28	"	"	"	"
E6	6C "	9,30	"	29	"	"	"	"
E7	7C "	32,33	"	12	"	"	"	"
E8	8C "	14,31	"	13	"	"	"	"
E9	9C "	15,34	"	35	"	"	"	"
E10	10C "	17,36	"	16	"	"	"	"
E11	11C "	19,38	"	40	"	"	"	"
E12	12C "	18,37	"	39	"	"	"	"

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TABLE VIII. Output voltage (high-level) test.

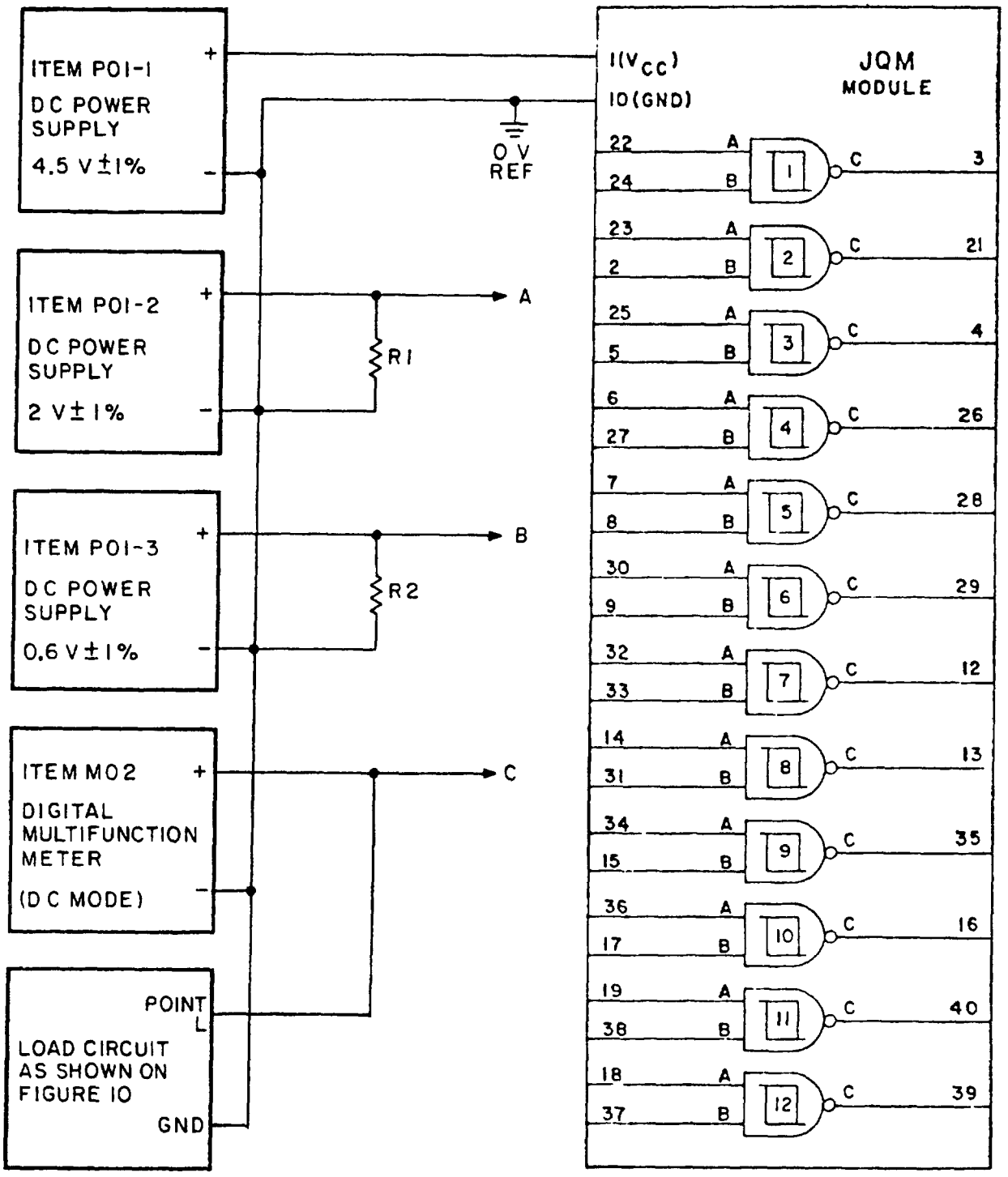
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1C Output voltage, high level	24	22	3	2.4	4.5	V	9
F2	1C "	22	24	3	"	"	"	"
F3	2C "	23	2	21	"	"	"	"
F4	2C "	2	23	21	"	"	"	"
F5	3C "	25	5	4	"	"	"	"
F6	3C "	5	25	4	"	"	"	"
F7	4C "	27	6	26	"	"	"	"
F8	4C "	6	27	26	"	"	"	"
F9	5C "	8	7	28	"	"	"	"
F10	5C "	7	8	28	"	"	"	"
F11	6C "	30	9	29	"	"	"	"
F12	6C "	9	30	29	"	"	"	"
F13	7C "	33	32	12	"	"	"	"
F14	7C "	32	33	12	"	"	"	"
F15	8C "	31	14	13	"	"	"	"
F16	8C "	14	31	13	"	"	"	"
F17	9C "	34	15	35	"	"	"	"
F18	9C "	15	34	35	"	"	"	"
F19	10C "	36	17	16	"	"	"	"
F20	10C "	17	36	16	"	"	"	"

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TABLE VIII. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F21	11C Output voltage, high level	38	19	40	2.4	4.5	V	9
F22	11C "	19	38	40	"	"	"	"
F23	12C "	37	18	39	"	"	"	"
F24	12C "	18	37	39	"	"	"	"

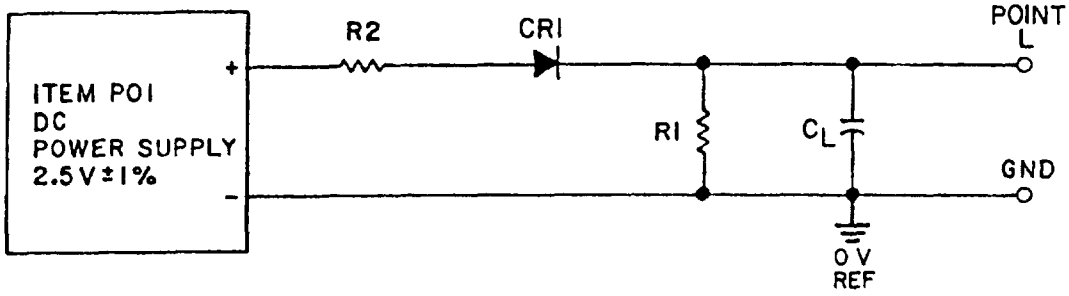
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NOTE R1 = R2 = 51Ω ±5%, 1 W.

* FIGURE 9. Output voltage (low-level and high-level) test circuit for tables VII and VIII.

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NOTES:
1. RI = Select value for current flow of 800 µA ±5% out of point "L" when "L" is held at 2.4 V.
2. R2 = Select value for current flow of 16 mA ±5% with point "L" held at 0.4 V.
3. CRI = 1N4150, or equivalent.
4. For test conditions P1-P12, Q1-Q12, R1-R12, S1-S12, T1-T24, and U1-U12, C_L = 50 ±5 pF (including probe and parasitic capacitance).
5. For test conditions P13-P24, Q13-Q24, R13-R24, and S13-S24, C_L = 150 ±5 pF (including probe and parasitic capacitance).
6. C_L is not required for test conditions E, F, K, L, or M.
7. The maximum lead length form point "L" to the appropriate module pin shall be less than 2 inches (50.8 mm).

* FIGURE 10. JQM logic load circuit.

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TABLE IX. Input current (low-level) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	1A	Input current, low level (V_{IN} = 0 V)	Open	22	Open	0.40	1.32	mA	11
G2	1B	"	"	24	"	"	"	"	"
G3	2A	"	"	23	"	"	"	"	"
G4	2B	"	"	2	"	"	"	"	"
G5	3A	"	"	25	"	"	"	"	"
G6	3B	"	"	5	"	"	"	"	"
G7	4A	"	"	6	"	"	"	"	"
G8	4B	"	"	27	"	"	"	"	"
G9	5A	"	"	7	"	"	"	"	"
G10	5B	"	"	8	"	"	"	"	"
G11	6A	"	"	30	"	"	"	"	"
G12	6B	"	"	9	"	"	"	"	"
G13	7A	"	"	32	"	"	"	"	"
G14	7B	"	"	33	"	"	"	"	"
G15	8A	"	"	14	"	"	"	"	"
G16	8B	"	"	31	"	"	"	"	"
G17	9A	"	"	34	"	"	"	"	"
G18	9B	"	"	15	"	"	"	"	"
G19	10A	"	"	36	"	"	"	"	"
G20	10B	"	"	17	"	"	"	"	"

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TABLE IX. Input current (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G21	11A Input current, low level (V_{IN} = 0 V)	Open	19	Open	0.40	1.32	mA	11
G22	11B "	"	38	"	"	"	"	"
G23	12A "	"	18	"	"	"	"	"
G24	12B "	"	37	"	"	"	"	"

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TABLE X. Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1B Input current, high level (V_{IN} = 2.4 V)	22	Open	24	0	40	µA	11
H2	1A "	24	"	22	"	"	"	"
H3	2A "	2	"	23	"	"	"	"
H4	2B "	23	"	2	"	"	"	"
H5	3A "	5	"	25	"	"	"	"
H6	3B "	25	"	5	"	"	"	"
H7	4B "	6	"	27	"	"	"	"
H8	4A "	27	"	6	"	"	"	"
H9	5B "	7	"	8	"	"	"	"
H10	5A "	8	"	7	"	"	"	"
H11	6A "	9	"	30	"	"	"	"
H12	6B "	30	"	9	"	"	"	"
H13	7B "	32	"	33	"	"	"	"
H14	7A "	33	"	32	"	"	"	"
H15	8B "	14	"	31	"	"	"	"
H16	8A "	31	"	14	"	"	"	"
H17	9A "	15	"	34	"	"	"	"
H18	9B "	34	"	15	"	"	"	"
H19	10A "	17	"	36	"	"	"	"
H20	10B "	36	"	17	"	"	"	"

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TABLE X. Input current (high-level) test - Continued.

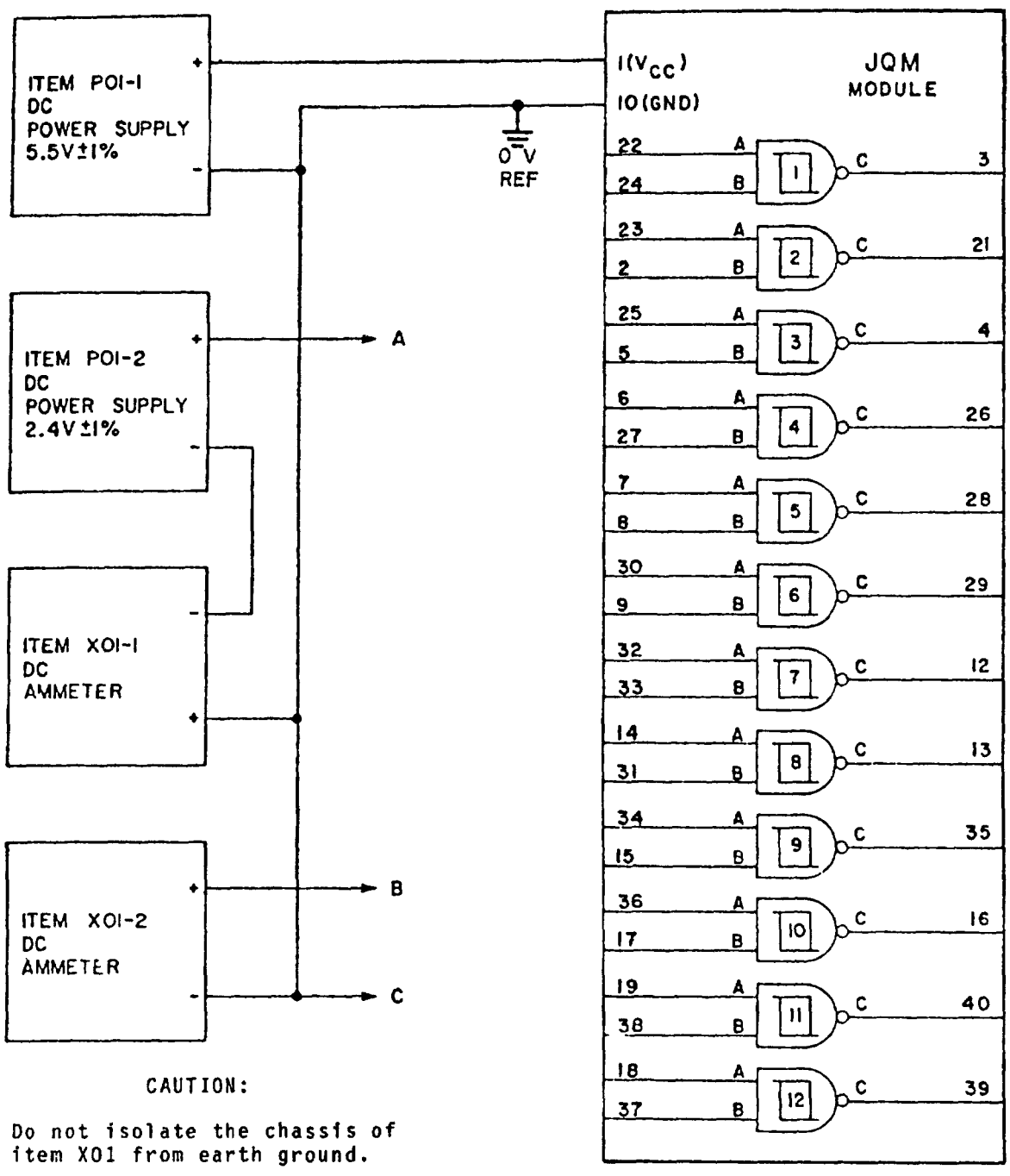
Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H21	11B	Input current, high level (V_{IN} = 2.4 V)	19	Open	38	0	40	µA	11
H22	11A	"	38	"	19	"	"	"	"
H23	12B	"	18	"	37	"	"	"	"
H24	12A	"	37	"	18	"	"	"	"

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TABLE XI. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1C Output short circuit current	Open	3	22	18	55	mA	11
J2	2C "	"	21	2	"	"	"	"
J3	3C "	"	4	5	"	"	"	"
J4	4C "	"	26	6	"	"	"	"
J5	5C "	"	28	7	"	"	"	"
J6	6C "	"	29	9	"	"	"	"
J7	7C "	"	12	32	"	"	"	"
J8	8C "	"	13	14	"	"	"	"
J9	9C "	"	35	15	"	"	"	"
J10	10C "	"	16	17	"	"	"	"
J11	11C "	"	40	19	"	"	"	"
J12	12C "	"	39	18	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTES:
1. Since items XO1-1 and XO1-2 are not used simultaneously, only one item XO1 is required.
2. With point A disconnected, item XO1-1 should indicate zero current on the 1 µA range.
3. When performing output short circuit current tests, only one output shall be shorted at a time and the measurement taken and the short removed within 5 seconds.

* FIGURE 11. Input current (low-level and high-level) and output short circuit current test circuit for tables IX, X, and XI.

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TABLE XII. Input threshold voltage (V_{THL} and V_{TLH}) and hysteresis (V_{THL} - V_{TLH}) test.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
K1	1A	Input threshold voltage (V_{THL})	24	22	3	1.5	2.0	V	12
K2	1B	"	22	24	3	"	"	"	"
K3	2A	"	2	23	21	"	"	"	"
K4	2B	"	23	2	21	"	"	"	"
K5	3A	"	5	25	4	"	"	"	"
K6	3B	"	25	5	4	"	"	"	"
K7	4A	"	27	6	26	"	"	"	"
K8	4B	"	6	27	26	"	"	"	"
K9	5A	"	8	7	28	"	"	"	"
K10	5B	"	7	8	28	"	"	"	"
K11	6A	"	9	30	29	"	"	"	"
K12	6B	"	30	9	29	"	"	"	"
K13	7A	"	33	32	12	"	"	"	"
K14	7B	"	32	33	12	"	"	"	"
K15	8A	"	31	14	13	"	"	"	"
K16	8B	"	14	31	13	"	"	"	"
K17	9A	"	15	34	35	"	"	"	"
K18	9B	"	34	15	35	"	"	"	"
K19	10A	"	17	36	16	"	"	"	"
K20	10B	"	36	17	16	"	"	"	"

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TABLE XII. Input threshold voltage (V_{THL} and V_{TLH}) and hysteresis (V_{THL} - V_{TLH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
K21	11A	Input threshold voltage (V_{THL})	38	19	40	1.5	2.0	V	12
K22	11B	"	19	38	40	"	"	"	"
K23	12A	"	37	18	39	"	"	"	"
K24	12B	"	18	37	39	"	"	"	"
L1	1A	Input threshold voltage (V_{TLH})	24	22	3	0.6	1.1	"	12
L2	1B	"	22	24	3	"	"	"	"
L3	2A	"	2	23	21	"	"	"	"
L4	2B	"	23	2	21	"	"	"	"
L5	3A	"	5	25	4	"	"	"	"
L6	3B	"	25	5	4	"	"	"	"
L7	4A	"	27	6	26	"	"	"	"
L8	4B	"	6	27	26	"	"	"	"
L9	5A	"	8	7	28	"	"	"	"
L10	5B	"	7	8	28	"	"	"	"
L11	6A	"	9	30	29	"	"	"	"
L12	6B	"	30	9	29	"	"	"	"
L13	7A	"	33	32	12	"	"	"	"
L14	7B	"	32	33	12	"	"	"	"
L15	8A	"	31	14	13	"	"	"	"
L16	8B	"	14	31	13	"	"	"	"

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TABLE XII. Input threshold voltage (V_{THL} and V_{TLH}) and hysteresis (V_{THL} - V_{TLH}) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
L17	9A	Input threshold voltage (V_{TLH})	15	34	35	0.6	1.1	V	12
L18	9B	"	34	15	35	"	"	"	"
L19	10A	"	17	36	16	"	"	"	"
L20	10B	"	36	17	16	"	"	"	"
L21	11A	"	38	19	40	"	"	"	"
L22	11B	"	19	38	40	"	"	"	"
L23	12A	"	37	18	39	"	"	"	"
L24	12B	"	18	37	39	"	"	"	"
M1	1A	Hysteresis (V_{THL} - V_{TLH})	24	22	3	0.4	1.4	"	12
M2	1B	"	22	24	3	"	"	"	"
M3	2A	"	2	23	21	"	"	"	"
M4	2B	"	23	2	21	"	"	"	"
M5	3A	"	5	25	4	"	"	"	"
M6	3B	"	25	5	4	"	"	"	"
M7	4A	"	27	6	26	"	"	"	"
M8	4B	"	6	27	26	"	"	"	"
M9	5A	"	8	7	28	"	"	"	"
M10	5B	"	7	8	28	"	"	"	"
M11	6A	"	9	30	29	"	"	"	"
M12	6B	"	30	9	29	"	"	"	"

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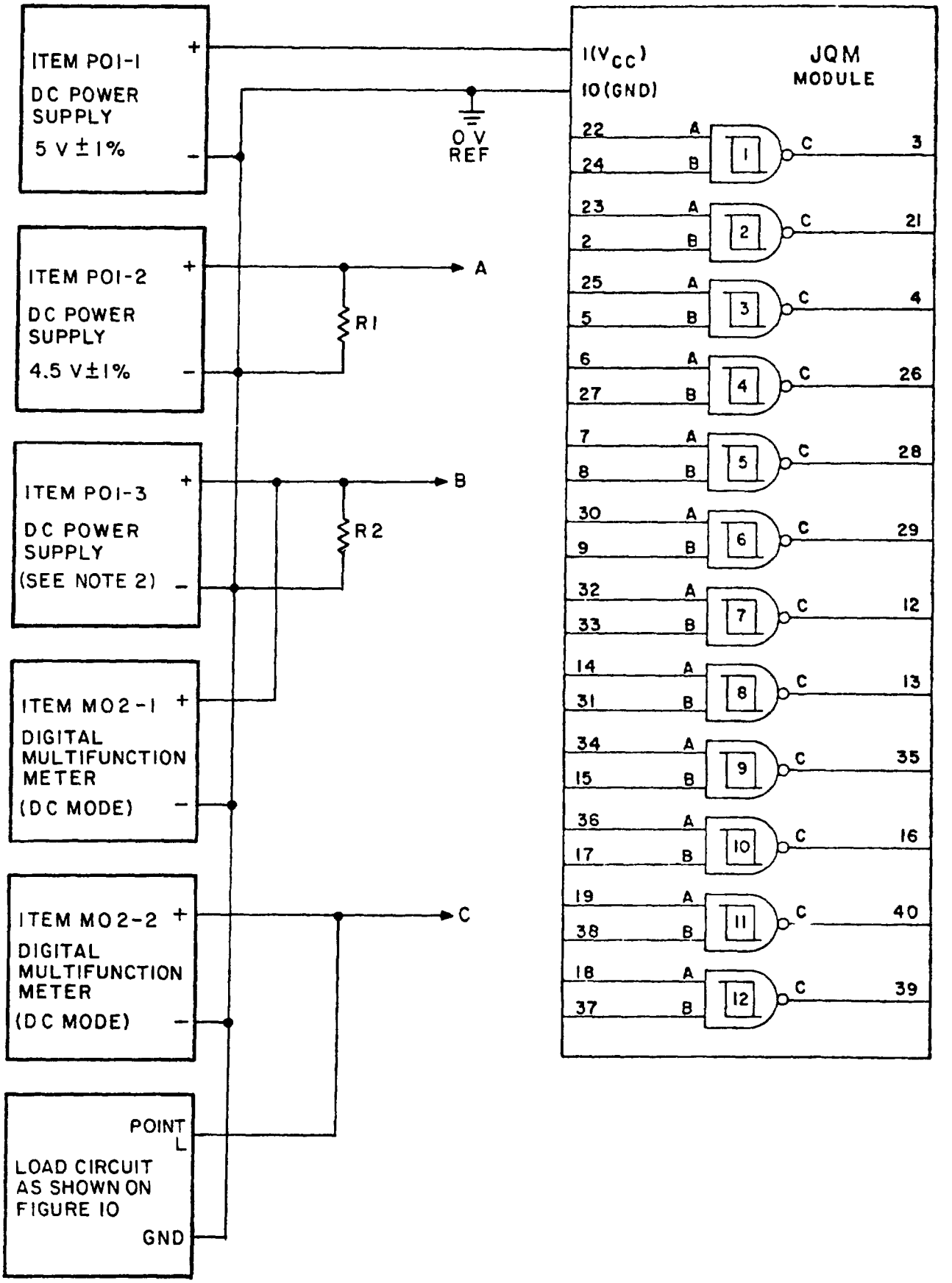
TABLE XII. Input threshold voltage (V_{THL} and V_{TLH}) and hysteresis (V_{THL} - V_{TLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M13	1A Hysteresis (V_{THL} - V_{TLH})	33	32	12	0.4	1.4	V	12
M14	7B "	32	33	12	"	"	"	"
M15	8A "	31	14	13	"	"	"	"
M16	8B "	14	31	13	"	"	"	"
M17	9A "	15	34	35	"	"	"	"
M18	9B "	34	15	35	"	"	"	"
M19	10A "	17	36	16	"	"	"	"
M20	10B "	36	17	16	"	"	"	"
M21	11A "	38	19	40	"	"	"	"
M22	11B "	19	38	40	"	"	"	"
M23	12A "	37	18	39	"	"	"	"
M24	12B "	18	37	39	"	"	"	"

TABLE XIII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	Power supply current, low level	C	Open	A	1	120	mA	13
N2	Power supply current, high level	B	A	Open	1	72	mA	13

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* FIGURE 12. Input threshold voltage (V_{THL} and V_{TLH}) and hysteresis (V_{THL} - V_{TLH}) test circuit for table XII.

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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Test procedure:

   (a) For each test condition, initially set item PO1-3 to 0 V.

   (b) Verify that the output of the gate under test is in a high-level state ($V_O \geq 2.4$ V).

   (c) Slowly increase the item PO1-3 output until the output of the gate under test switches logic states and goes to a low-level ($V_O \leq 0.4$ V). Record the input voltage (V_{THL}) as test condition K.

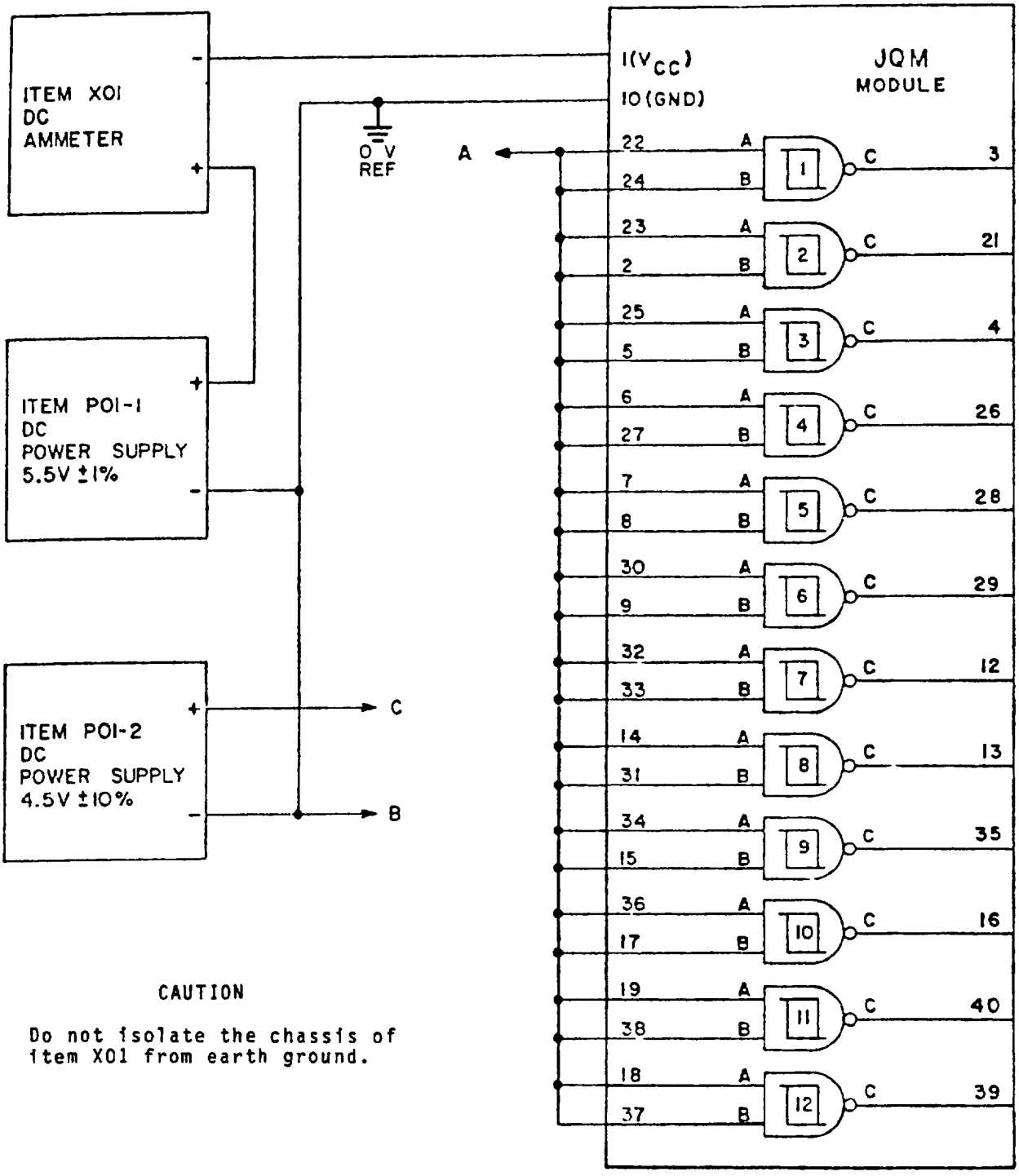
   (d) Slowly decrease the item PO1-3 output voltage until the output of the gate under test switches logic states and goes to a high-level ($V_O \geq 2.4$ V). Record the input voltage (V_{TLH}) as test condition L.

   (e) Test condition M is a calculation obtained by subtracting V_{TLH} input voltage (test condition L) from V_{THL} input voltage (test condition K). Record ($V_{THL} - V_{TLH}$) as test condition M.

   (f) Decrease item PO1-3 voltage to 0 V and repeat test procedure for each test condition.

* FIGURE 12. Input threshold voltage (V_{THL} and V_{TLH}) and hysteresis ($V_{THL} - V_{TLH}$) test circuit for table XII - Continued.

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* FIGURE 13. Power supply current (low-level and high-level) test circuit for table XIII.

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	1C Transition time (t_{TLH}), C_L = 50 pF	22	24	3	4	45	ns	14,15
P2	2C "	2	23	21	"	"	"	"
P3	3C "	5	25	4	"	"	"	"
P4	4C "	6	27	26	"	"	"	"
P5	5C "	7	8	28	"	"	"	"
P6	6C "	9	30	29	"	"	"	"
P7	7C "	32	33	12	"	"	"	"
P8	8C "	14	31	13	"	"	"	"
P9	9C "	15	34	35	"	"	"	"
P10	10C "	17	36	16	"	"	"	"
P11	11C "	19	38	40	"	"	"	"
P12	12C "	18	37	39	"	"	"	"
P13	1C Transition time (t_{TLH}), C_L = 150 pF	22	24	3	"	55	"	"
P14	2C "	2	23	21	"	"	"	"
P15	3C "	5	25	4	"	"	"	"
P16	4C "	6	27	26	"	"	"	"
P17	5C "	7	8	28	"	"	"	"
P18	6C "	9	30	29	"	"	"	"
P19	7C "	32	33	12	"	"	"	"
P20	8C "	14	31	13	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P21	9C	Transition time (t_{TLH}), C_L = 150 pF	15	34	35	4	55	ns	14,15
P22	10C	"	17	36	16	"	"	"	"
P23	11C	"	19	38	40	"	"	"	"
P24	12C	"	18	37	39	"	"	"	"
Q1	1C	Transition time (t_{THL}), C_L = 50 pF	22	24	3	"	20	"	"
Q2	2C	"	2	23	21	"	"	"	"
Q3	3C	"	5	25	4	"	"	"	"
Q4	4C	"	6	27	26	"	"	"	"
Q5	5C	"	7	8	28	"	"	"	"
Q6	6C	"	9	30	29	"	"	"	"
Q7	7C	"	32	33	12	"	"	"	"
Q8	8C	"	14	31	13	"	"	"	"
Q9	9C	"	15	34	35	"	"	"	"
Q10	10C	"	17	36	16	"	"	"	"
Q11	11C	"	19	38	40	"	"	"	"
Q12	12C	"	18	37	39	"	"	"	"
Q13	1C	Transition time (t_{THL}), C_L = 150 pF	22	24	3	"	30	"	"
Q14	2C	"	2	23	21	"	"	"	"
Q15	3C	"	5	25	4	"	"	"	"
Q16	4C	"	6	27	26	"	"	"	"
Q17	5C	"	7	8	28	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q18	6C	Transition time (t_{THL}), C_L = 150 pF	9	30	29	4	30	ns	14,15
Q19	7C	"	32	33	12	"	"	"	"
Q20	8C	"	14	31	13	"	"	"	"
Q21	9C	"	15	34	35	"	"	"	"
Q22	10C	"	17	36	16	"	"	"	"
Q23	11C	"	19	38	40	"	"	"	"
Q24	12C	"	18	37	39	"	"	"	"

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TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	1C Propagation delay time (t_{PHL}), C_L = 50 pF	22	24	3	5	30	ns	14,15
R2	2C "	2	23	21	"	"	"	"
R3	3C "	5	25	4	"	"	"	"
R4	4C "	6	27	26	"	"	"	"
R5	5C "	7	8	28	"	"	"	"
R6	6C "	9	30	29	"	"	"	"
R7	7C "	32	33	12	"	"	"	"
R8	8C "	14	31	13	"	"	"	"
R9	9C "	15	34	35	"	"	"	"
R10	10C "	17	36	16	"	"	"	"
R11	11C "	19	38	40	"	"	"	"
R12	12C "	18	37	39	"	"	"	"
R13	1C Propagation delay time (t_{PHL}), C_L = 150 pF	22	24	3	"	40	"	"
R14	2C "	2	23	21	"	"	"	"
R15	3C "	5	25	4	"	"	"	"
R16	4C "	6	27	26	"	"	"	"
R17	5C "	7	8	28	"	"	"	"
R18	6C "	9	30	29	"	"	"	"
R19	7C "	32	33	12	"	"	"	"
R20	8C "	14	31	13	"	"	"	"

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TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

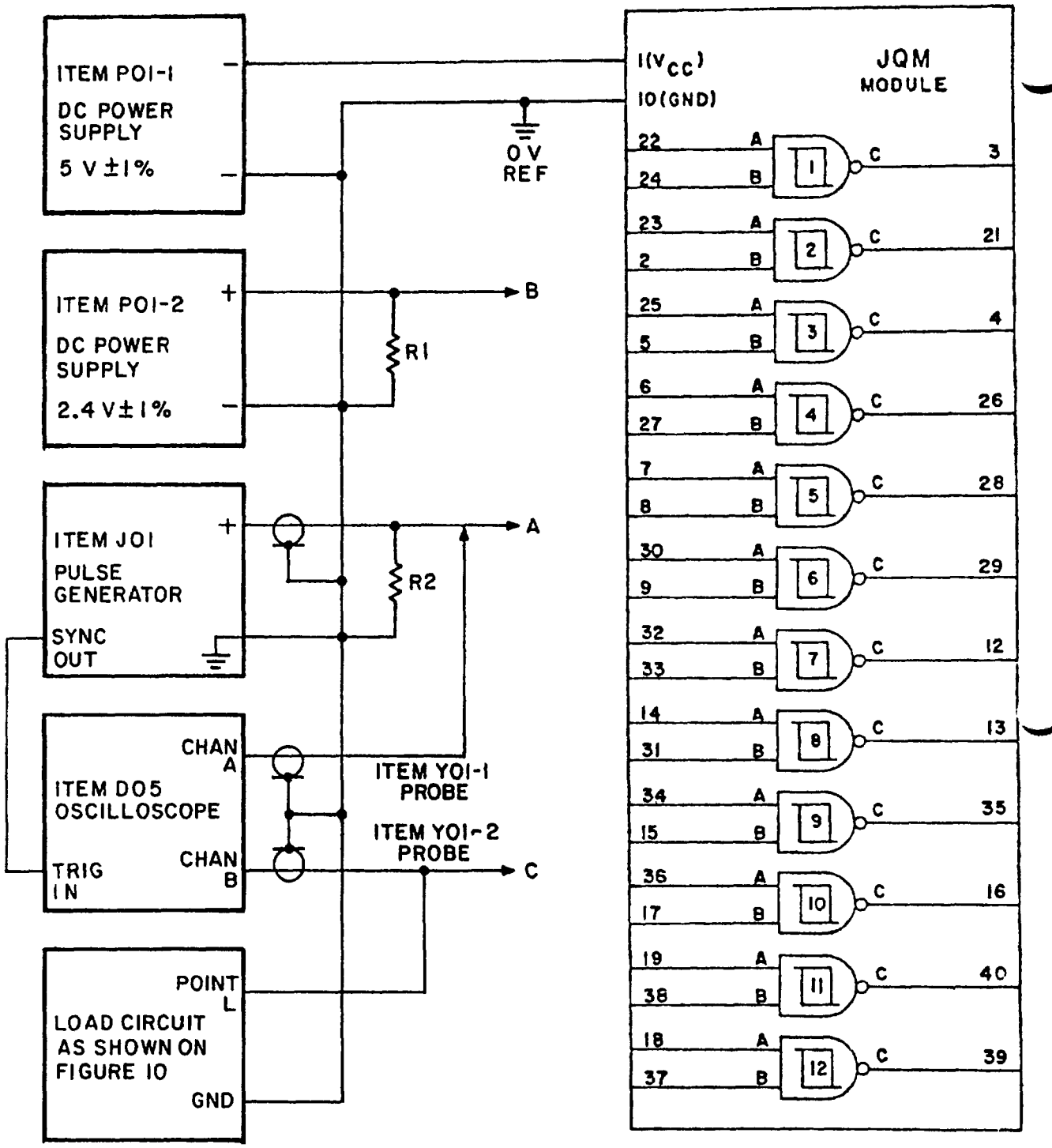
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R21	9C Propagation delay time (t_{PHL}), C_L = 150 pF	15	34	35	5	40	ns	14,15
R22	10C "	17	36	16	"	"	"	"
R23	11C "	19	38	40	"	"	"	"
R24	12C "	18	37	39	"	"	"	"
S1	1C Propagation delay time (t_{PLH}), C_L = 50 pF	22	24	3	"	30	"	"
S2	2C "	2	23	21	"	"	"	"
S3	3C "	5	25	4	"	"	"	"
S4	4C "	6	27	26	"	"	"	"
S5	5C "	7	8	28	"	"	"	"
S6	6C "	9	30	29	"	"	"	"
S7	7C "	32	33	12	"	"	"	"
S8	8C "	14	31	13	"	"	"	"
S9	9C "	15	34	35	"	"	"	"
S10	10C "	17	36	16	"	"	"	"
S11	11C "	19	38	40	"	"	"	"
S12	12C "	18	37	39	"	"	"	"
S13	1C Propagation delay time, (t_{PLH}), C_L = 150 pF	22	24	3	"	40	"	"
S14	2C "	2	23	21	"	"	"	"
S15	3C "	5	25	4	"	"	"	"
S16	4C "	6	27	26	"	"	"	"

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TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition		Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
S17	5C	Propagation delay time, (t_{PLH}), C_L = 150 pF	7	8	28	5	40	ns	14,15
S18	6C	"	9	30	29	"	"	"	"
S19	7C	"	32	33	12	"	"	"	"
S20	8C	"	14	31	13	"	"	"	"
S21	9C	"	15	34	35	"	"	"	"
S22	10C	"	17	36	16	"	"	"	"
S23	11C	"	19	38	40	"	"	"	"
S24	12C	"	18	37	39	"	"	"	"

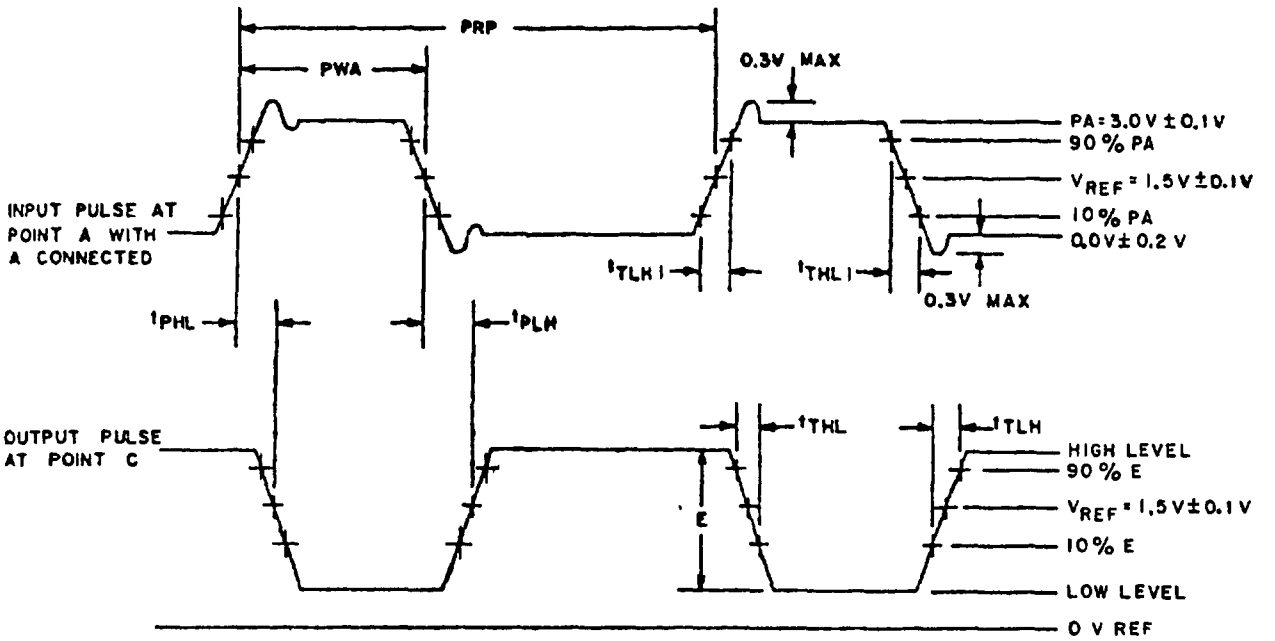
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Before making these measurements, the output levels shall be verified to be ≤ 0.4 V for the low level and ≥ 2.4 V for the high level.

* FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XIV and XV.

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NOTES.
1. Input pulse parameters:
   t_{TLH} = t_{THL} = 10 ±2 ns.
   PWA = 400 ±40 ns.
   PRP = 1,000 ±100 ns.
2. All voltages are referenced to ground.

* FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XIV and XV.

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TABLE XVI. Output crosstalk (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1C Output crosstalk, low level	Open	E	F,G,H,J,K,L, M,N,P,R,S	3	0	1	V	16,17
T2	2C "	"	F	E,G,H,J,K,L, M,N,P,R,S	21	"	"	"	"
T3	3C "	"	G	E,F,H,J,K,L, M,N,P,R,S	4	"	"	"	"
T4	4C "	"	H	E,F,G,J,K,L, M,N,P,R,S	26	"	"	"	"
T5	5C "	"	J	E,F,G,H,K,L, M,N,P,R,S	28	"	"	"	"
T6	6C "	"	K	E,F,G,H,J,L, M,N,P,R,S	29	"	"	"	"
T7	7C "	"	L	E,F,G,H,J,K, M,N,P,R,S	12	"	"	"	"
T8	8C "	"	M	E,F,G,H,J,K, L,N,P,R,S	13	"	"	"	"
T9	9C "	"	N	E,F,G,H,J,K, L,M,P,R,S	35	"	"	"	"
T10	10C "	"	P	E,F,G,H,J,K, L,M,N,R,S	16	"	"	"	"
T11	11C "	"	R	E,F,G,H,J,K, L,M,N,P,S	40	"	"	"	"
T12	12C "	"	S	E,F,G,H,J,K, L,M,N,P,R	39	"	"	"	"

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TABLE XVI. Output crosstalk (low-level) test - Continued.

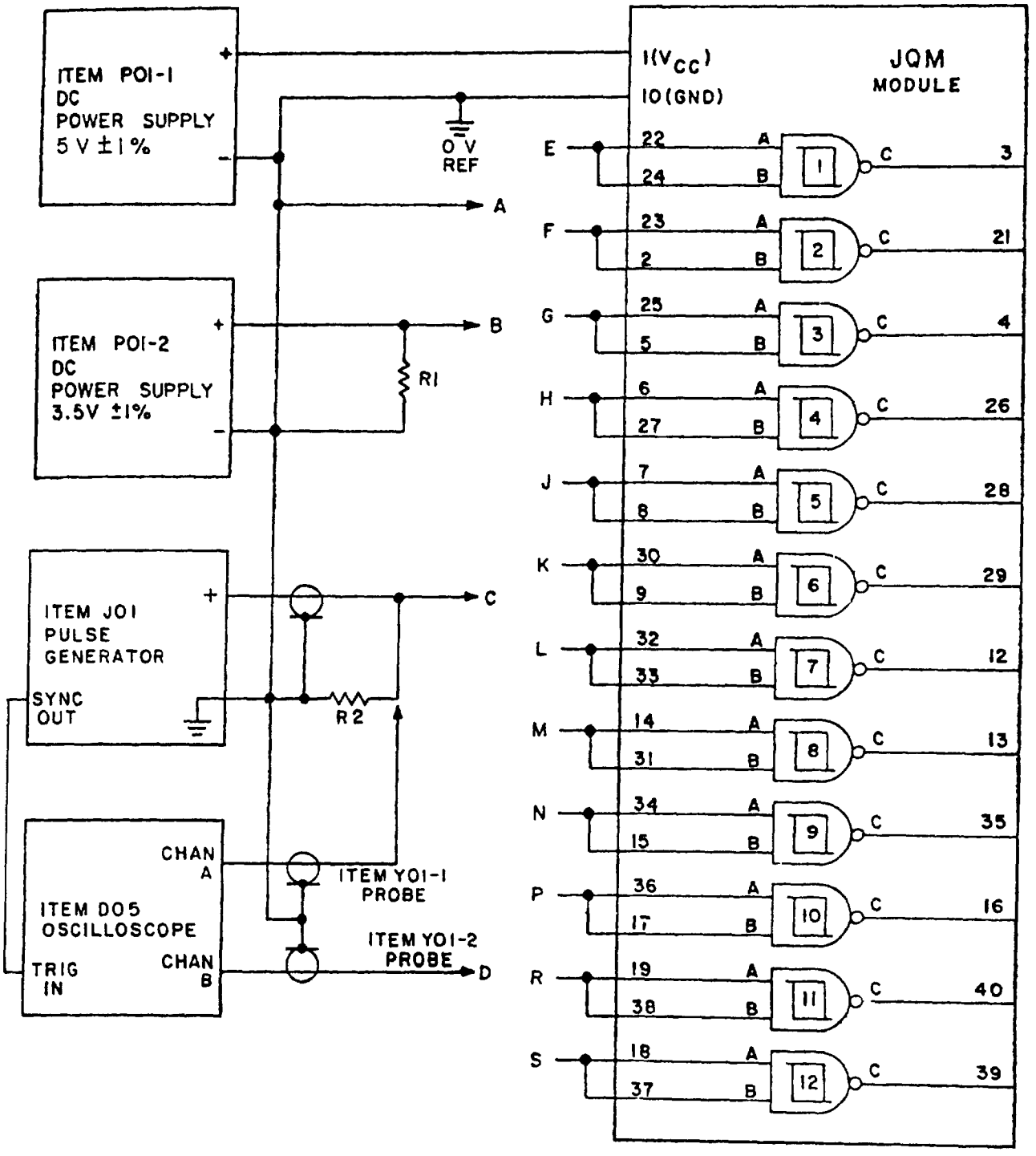
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T13	1C Output crosstalk, low level	Open	E,F,G,H	J,K,L,M,N,P, R,S	3	0	400	mV	16,17
T14	2C "	"	"	"	21	"	"	"	"
T15	3C "	"	"	"	4	"	"	"	"
T16	4C "	"	"	"	26	"	"	"	"
T17	5C "	"	J,K,L,M	E,F,G,H,N,P, R,S	28	"	"	"	"
T18	6C "	"	"	"	29	"	"	"	"
T19	7C "	"	"	"	12	"	"	"	"
T20	8C "	"	"	"	13	"	"	"	"
T21	9C "	"	N,P,R,S	E,F,G,H,J,K, L,M	35	"	"	"	"
T22	10C "	"	"	"	16	"	"	"	"
T23	11C "	"	"	"	40	"	"	"	"
T24	12C "	"	"	"	39	"	"	"	"

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TABLE XVII. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1C Output crosstalk, high level	E	Open	F,G,H,J,K,L, M,N,P,R,S	3	0	400	mV	16,17
U2	2C "	F	"	E,G,H,J,K,L, M,N,P,R,S	21	"	"	"	"
U3	3C "	G	"	E,F,H,J,K,L, M,N,P,R,S	4	"	"	"	"
U4	4C "	H	"	E,F,G,J,K,L, M,N,P,R,S	26	"	"	"	"
U5	5C "	J	"	E,F,G,H,K,L, M,N,P,R,S	28	"	"	"	"
U6	6C "	K	"	E,F,G,H,J,L, M,N,P,R,S	29	"	"	"	"
U7	7C "	L	"	E,F,G,H,J,K, M,N,P,R,S	12	"	"	"	"
U8	8C "	M	"	E,F,G,H,J,K, L,N,P,R,S	13	"	"	"	"
U9	9C "	N	"	E,F,G,H,J,K, L,M,P,R,S	35	"	"	"	"
U10	10C "	P	"	E,F,G,H,J,K, L,M,N,R,S	16	"	"	"	"
U11	11C "	R	"	E,F,G,H,J,K, L,M,N,P,S	40	"	"	"	"
U12	12C "	S	"	E,F,G,H,J,K, L,M,N,P,R	39	"	"	"	"

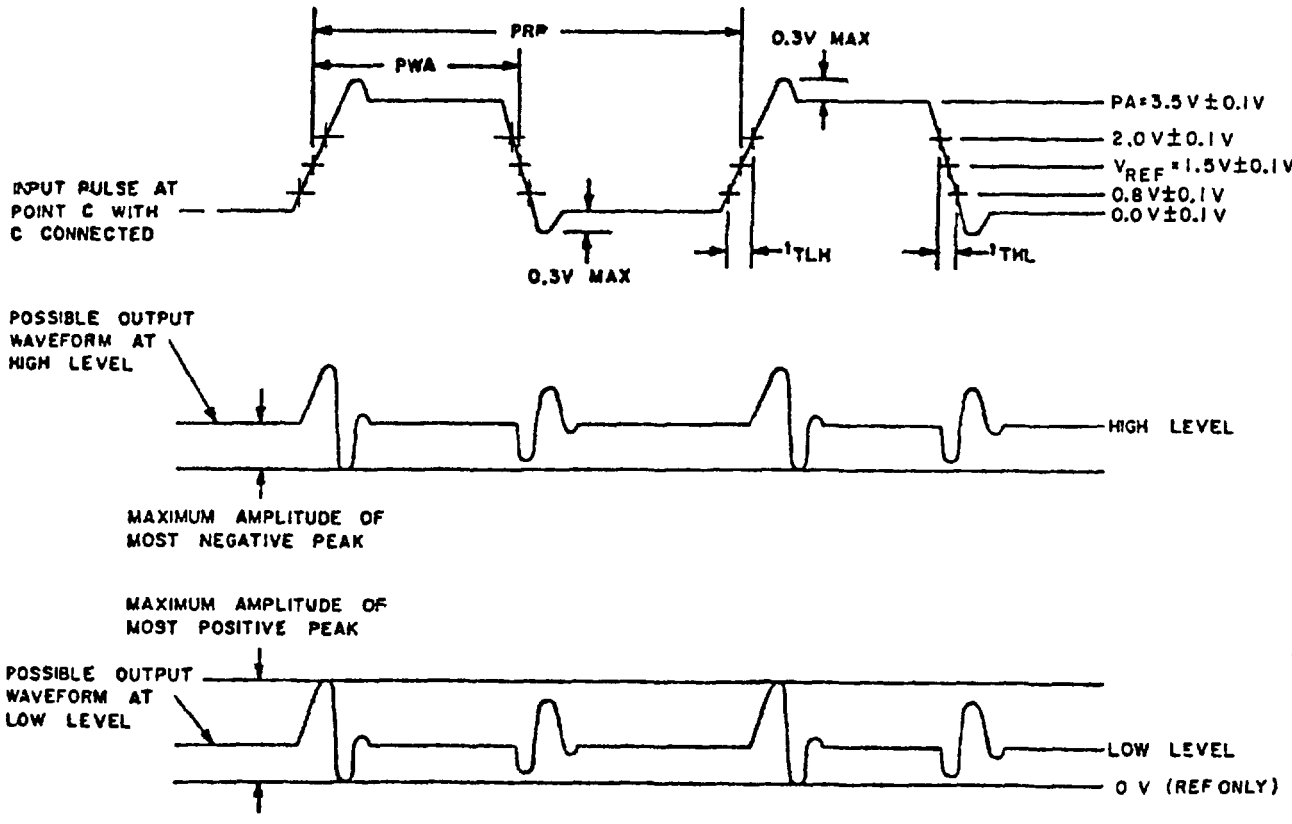
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1W.
2. Load each output with a load as shown on figure 10.

* FIGURE 16 Output crosstalk (low-level and high-level) test circuit for tables XVI and XVII.

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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH}$ = 3.5 ±1.5 ns (verified at each input pin).
   PWA = 250 ±30 ns.
   PRP = 500 ±50 ns.
2. All voltages are referenced to ground.

FIGURE 17. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVI and XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
V	Power supply transient amplitude	0	400	mV	18,19

TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W	Power supply filter capacitance	Open	1	1.6	3.0	µF	20

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TABLE XX. Input capacitance test.

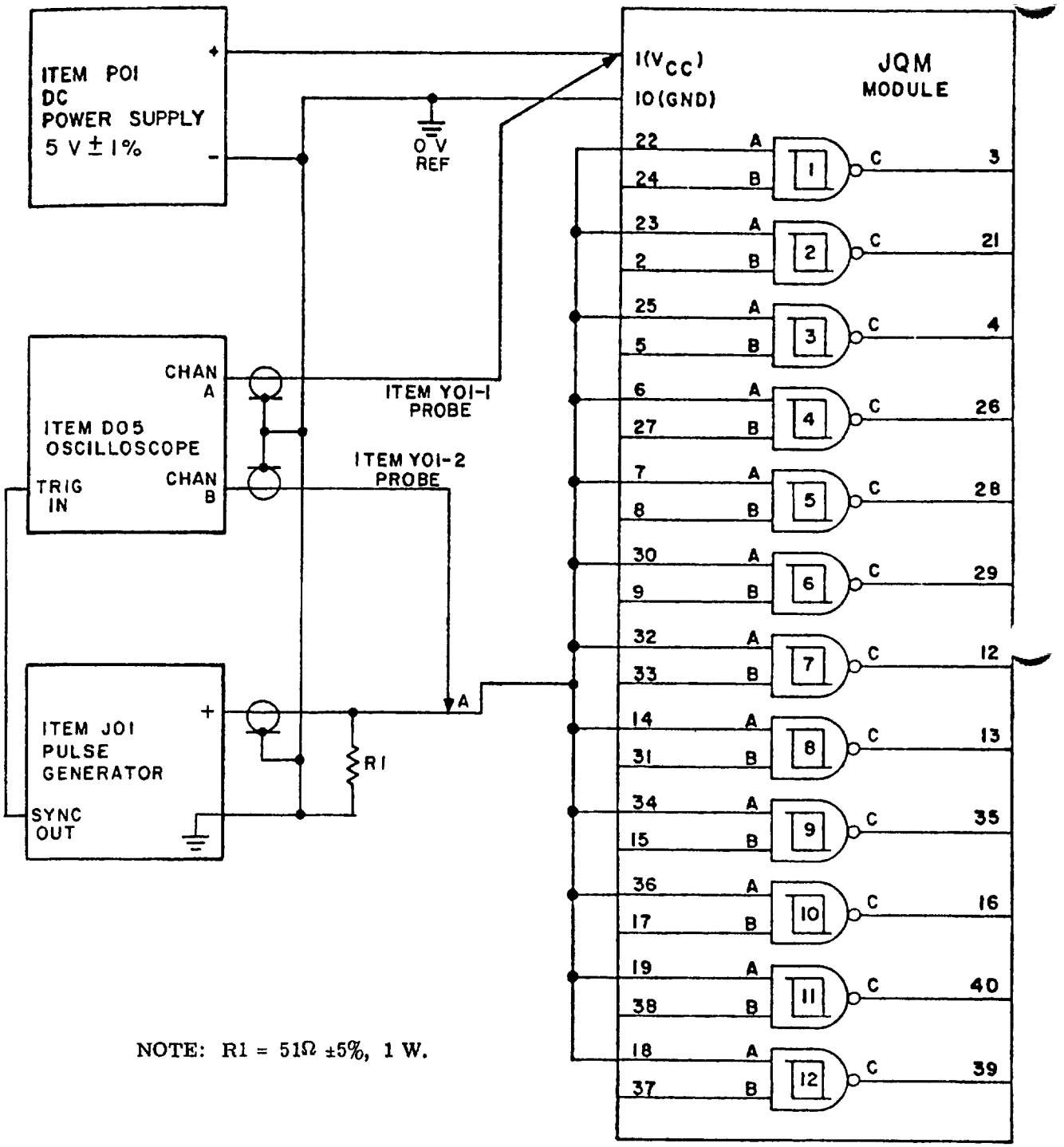
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X1	1A Input capacitance	22	Open	2	15	pF	20
X2	1B "	24	"	"	"	"	"
X3	2B "	2	"	"	"	"	"
X4	2A "	23	"	"	"	"	"
X5	3B "	5	"	"	"	"	"
X6	3A "	25	"	"	"	"	"
X7	4A "	6	"	"	"	"	"
X8	4B "	27	"	"	"	"	"
X9	5A "	7	"	"	"	"	"
X10	5B "	8	"	"	"	"	"
X11	6B "	9	"	"	"	"	"
X12	6A "	30	"	"	"	"	"
X13	7A "	32	"	"	"	"	"
X14	7B "	33	"	"	"	"	"
X15	8A "	14	"	"	"	"	"
X16	8B "	31	"	"	"	"	"
X17	9B "	15	"	"	"	"	"
X18	9A "	34	"	"	"	"	"
X19	10B "	17	"	"	"	"	"
X20	10A "	36	"	"	"	"	"

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TABLE XX. Input capacitance test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
X21	11A	Input capacitance	19	Open	2	15	pF	20
X22	11B	"	38	"	"	"	"	"
X23	12A	"	18	"	"	"	"	"
X24	12B	"	37	"	"	"	"	"

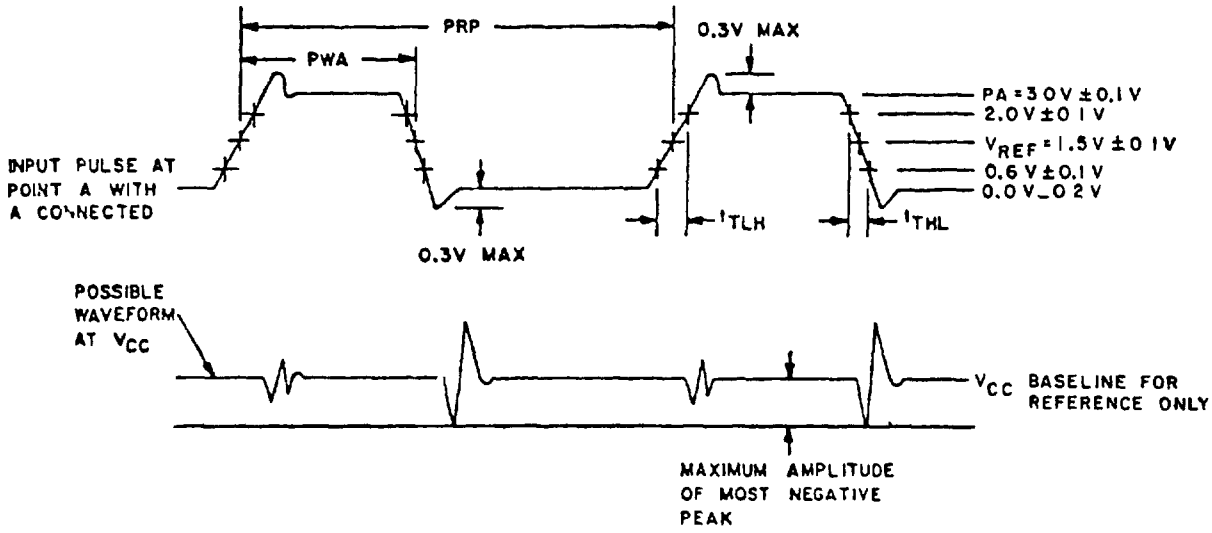
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NOTE: R1 = 51Ω ±5%, 1 W.

* FIGURE 18. Power supply transient amplitude test circuit for table XVIII.

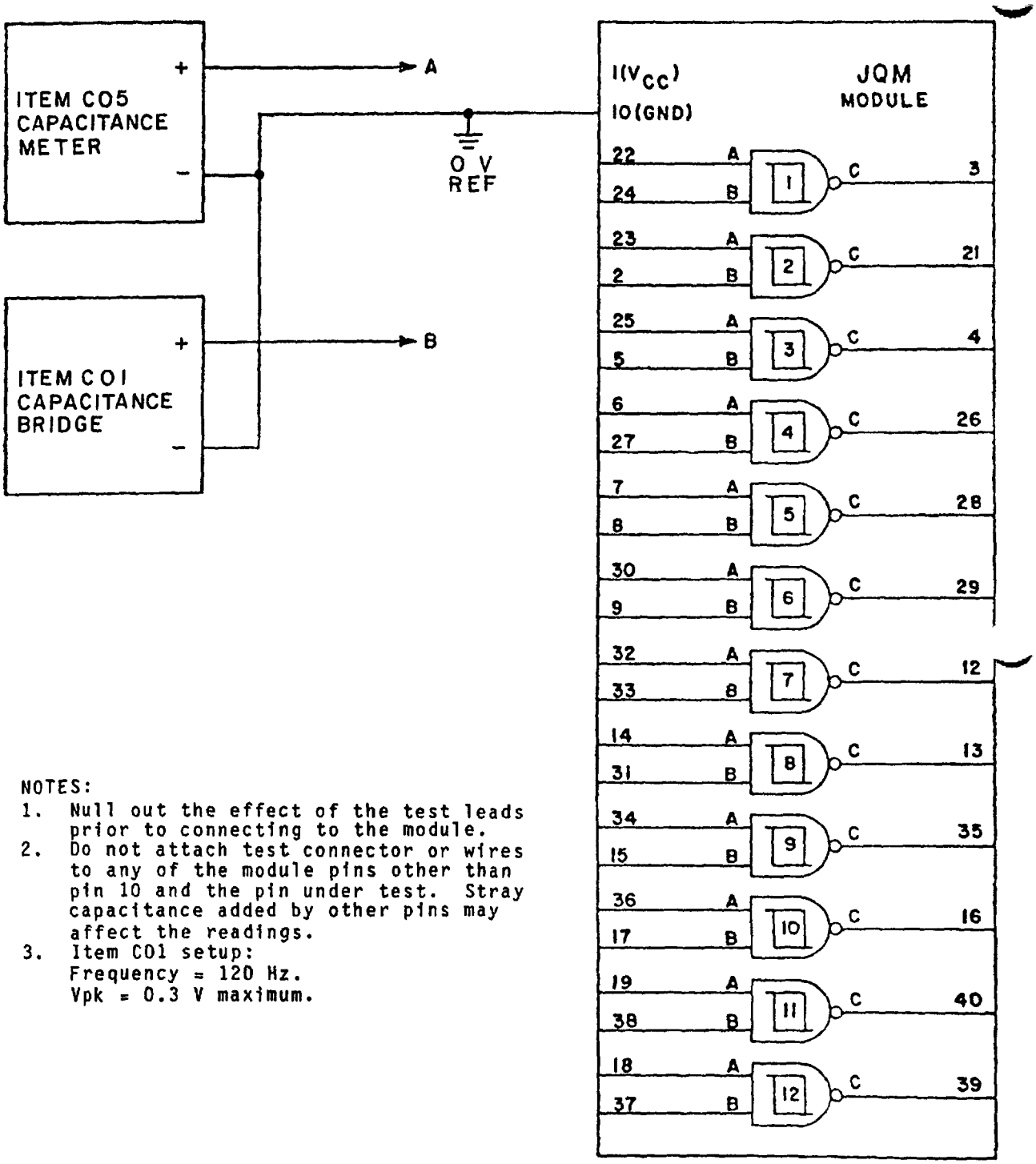
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NOTES:
1. Input pulse parameters.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. All voltages are referenced to ground.

* FIGURE 19. Power supply transient amplitude test circuit waveforms for table XVIII.

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added by other pins may affect the readings.
3. Item C01 setup:
   Frequency = 120 Hz.
   Vpk = 0.3 V maximum.

* FIGURE 20. Power supply filter capacitance and input capacitance test circuit for tables XIX and XX.

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DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20363
ATTN: ELEX 8111

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/222A(EC)	Modules, Standard Electronic Nand/Schmitt, Digital, Key Code JQM

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6. REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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